Key Terms, Weeks 1–7

Every study-guide entry so far, in a line or two. Use it to find what you cannot yet define; then read the full entry, which also says what you should be able to *do* with the idea.

Week 1 design vocabulary

Affordance — what a designed thing makes possible for the person using it.

Constraint — what it prevents, on purpose. A prompt is where you add the constraints a chat box leaves out.

Signifier — the visible clue that an affordance is there at all.

Week 2 how text becomes numbers

THE MACHINERY WORDS

Corpus — the body of text a method learns from during training. Afterwards only its effects remain.

Vocabulary — the fixed list of items a method can represent. Anything else is out of vocabulary.

Vector — an ordered list of numbers of fixed length: a point in a space, and an arrow with a direction.

Dimensionality — how many numbers are in each vector.

Neural network, layer — a stack of layers, each turning one list of numbers into another. “Deep” means many layers.

Parameters (weights) — the numbers inside the layers that training adjusts. What “open weights” releases.

Training / inference — training sets the parameters once from a corpus; inference is every use of the finished model.

Softmax — turns any list of scores into probabilities between 0 and 1 that add up to 1.

TEXT INTO NUMBERS

Token, tokenization — cutting text into subword pieces from the model's vocabulary. A token is not a word.

One-hot encoding — a vector as long as the vocabulary with a single 1 at the word's position.

Sparse / dense vector — sparse: long and mostly zeros (one-hot, counts, TF-IDF). Dense: short, every entry used (embeddings).

Embedding — a token turned into a dense vector whose numbers carry something about meaning.

THE GEOMETRY

Embedding space — the shared space one model's embeddings live in, where similar meaning becomes “close together”.

Dot product — multiply two vectors entry by entry and add. Large when they point the same way; also grows with length.

Cosine similarity — the dot product divided by both lengths, leaving only the angle: 1 same direction, 0 unrelated, −1 opposite.

Nearest neighbors — the k vectors closest to a given one, usually by cosine similarity.

Dimensionality reduction — squashing vectors to two or three dimensions so they can be drawn (PCA, t-SNE, UMAP).

Vector arithmetic — differences between vectors encode relationships (king − man + woman ≈ queen), with real limits.

COUNTING METHODS

Bag of words — a document as the words it contains and how often, with order discarded.

Count vectorization — one column per term, one row per document, counts in the cells: a document-term matrix.

Stop words — very common words (“the”, “of”) removed before counting.

Stemming — chopping endings off by rule. Fast; the result need not be a real word.

Lemmatization — reducing a word to its dictionary form using a vocabulary. Always a real word.

N-grams — runs of adjacent words counted as units: bigrams (two), trigrams (three).

Term frequency (TF) — how often a term appears in one document, divided by the document's length.

Inverse document frequency (IDF) — how rare a term is across the corpus. Zero for a term in every document.

TF-IDF — TF × IDF: high when a term is frequent here and rare elsewhere.

LEARNED EMBEDDINGS

Distributional hypothesis — words that appear in similar contexts have similar meanings. A bet, not a theorem.

Sliding window — a fixed-width window moved along the text; each stop is one training example.

CBOW — predict the middle word from its neighbors. Context in, one word out.

Skip-gram — predict the neighbors from the middle word. One word in, context out.

Negative sampling — train against a few random wrong words instead of a softmax over the whole vocabulary.

word2vec — the 2013 model that learns one dense vector per word with CBOW or skip-gram.

Pretext task — a training task nobody wants the answers to, set so the network must learn a useful representation.

CONTEXT AND SEQUENCES

Polysemy — one word with several meanings (“bank”). Ordinary, not exotic.

Encoder / decoder — an encoder reads a sequence into a representation; a decoder writes a sequence from one.

Pretraining / fine-tuning — train once on a huge general corpus, then continue on a small task-specific one.

Masked language modeling — BERT's task: hide some words and recover them using the words on both sides.

BERT — an encoder-only transformer (2018) that reads text and produces representations.

Static / contextual embedding — static: one fixed vector per word (word2vec). Contextual: computed per occurrence from the sentence.

Recurrent neural network (RNN) — reads one item at a time, updating a running summary.

Hidden state — that running summary: a fixed-size vector, however much has been read.

Sequence-to-sequence — an encoder and a decoder joined only by the encoder's final hidden state, the context vector.

Fixed-length bottleneck — the whole input squeezed into that one vector. The problem attention solved.

THE OUTPUT END

Logits — the raw scores a model produces, one per token in the vocabulary, before softmax.

Temperature — a number the logits are divided by. Higher flattens the choice; zero always takes the top token.

Week 3 attention and the transformer

Architecture / model / technique — a design (the transformer); one trained instance of it (BERT); a method used on or around a model (a prompt).

Attention (self-attention) — every word looks at every other word in one step and is adjusted by what it finds.

Query — the question a word asks of the others.

Key — the label a word wears, matched against other words' queries.

Value — what a word hands over to any word that attends to it.

Attention score — the dot product of one word's query with another's key.

Attention weights — a row of scores after softmax: shares that add up to 1. Attention is a budget.

Multi-head attention — several heads side by side in a layer, each with its own learned questions.

Positional encoding — a vector marking a token's position, added to its embedding so word order is not lost.

Causal mask — scores for later words set to minus infinity before softmax, so a decoder cannot read ahead. Weights are exactly zero.

Cross-attention — queries from the decoder, keys and values from the encoder: how the writing half looks at what was read.

Transformer block — attention, then a feed-forward network, with "add and normalize" after each. Stacked many times.

Add and normalize — the block's input is added back to its output, so nothing is fully overwritten.

What the transformer traded — everything at once (parallel training) for cost that grows with the square of the length. Generation is still one token at a time.

Attention is not explanation — a weight shows how one head blended values, not why the model produced its output.

Week 6 chain of thought and its critics

Zero-shot / few-shot — a prompt with no worked examples, or with one or more (exemplars) before the question.

Chain-of-thought prompting — asking for intermediate steps before the final answer.

Benchmark — a fixed set of questions with known answers that everyone scores on.

Emergent ability — absent in small models, appearing above some size.

Ablation study — remove or replace one part at a time and compare with the full version (the control).

Training / test distribution — the pattern the training examples follow, and the pattern of the held-back examples used to check.

Out-of-distribution (OOD) — questions outside the pattern of the training data. Doing well on them is OOD generalization.

Data leakage — test questions or answers turning up in training data, so memory cannot be told from method.

Exact match — right only if identical to the expected answer.

Edit distance — the fewest single-character changes that turn one string into another. Lower is better.

BLEU — a score for how many short runs two texts share. 1 is identical.

Faithfulness — whether the steps a model writes are how it actually reached its answer.

Loss / risk — the penalty for one answer; the average penalty. Empirical risk is over the questions seen, expected risk over all a distribution could produce.

Total variation distance — how different two distributions are: 0 identical, 1 no overlap.

Generalization bound — a guarantee that test risk is at most training risk plus stated costs. A ceiling, not a prediction.

Week 7 truthfulness and evaluation

Hallucination — a fluent, confident statement that is false or cites something that does not exist.

Imitative falsehood — a false answer given because the model learned its training text well: the popular myth is the likely continuation.

Inverse scaling — a result that gets worse as models get larger, as truthfulness did on TruthfulQA in 2021.

Truthful — avoids asserting anything false. "I have no comment" is truthful.

Informative — actually addresses what was asked. The number that matters is answers that are both.

Ground truth — the answer an evaluation treats as correct: checked, sourced, and written by a person.

Evaluation set — a fixed list of inputs with ground truth that every version of a prompt is run against.

Marking rule — the rule that turns an answer and the ground truth into right or wrong. Part of your definition of correct.

Inter-annotator agreement — how far two people marking the same answers give the same marks.

Cohen's kappa — agreement corrected for chance: 1 perfect, 0 no better than chance.

Small-sample uncertainty — with twenty items, one item is five points and any score has a wide plausible range.

How to use this page. Cover the right-hand side of a line and define the term out loud. If you can, go on. If you cannot, mark it and read the full study-guide entry.

Knowing a definition is the start. Concept checks ask you to *apply* the idea, so for each marked term also try one of the "you should be able to" lines in its entry.